Optimization of Traffic System Based on Ant Colony Algorithm and Dijkstra's Algorithm

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***Abstract*—This study focuses on the urban transportation system by using the ant colony algorithm and Dijkstra's algorithm, aiming to comprehensively analyze the transportation system by combining various models. First, in the data processing stage, the median padding method and IQR outlier identification method are used to deal with the missing values and outliers, and the maximum normalization is used to standardize the AADT data in order to improve the data quality. Next, a weighted directed graph model is used to describe the transportation system, and Dijkstra's algorithm is used to evaluate the impact of key events on transportation. Next, the ACO algorithm is used to optimize the bus routes and construct the weighted directed graph. Finally, through the hierarchical analysis method (AHP) combined with the entropy method, the optimized public transportation system is determined as the optimal solution by considering multiple factors. The advantage of this study is that it provides a comprehensive and in-depth analysis for the optimization of the transportation system and a scientific basis for urban transportation planning and decision-making.**

Keywords—traffic network modeling, weighted directed graph model, Ant Colony Algorithm (ACO), Dijkstra's algorithm, Hierarchical Analysis (AHP)

I. Introduction

In this paper, a combination of computer-related algorithms and models are used to address urban traffic situations[1]. The research covers analyzing urban traffic, assessing the impacts of various projects on the transportation system and stakeholders, and exploring how to ensure safety through the transportation system. Firstly, traffic network modeling[2] and directed graphical modeling[3] are used to describe the traffic system, and Dijkstra's algorithm[4] is applied to assess the impact of bridge collapse on traffic. Next, the ant colony algorithm (ACO) is used to optimize the bus routes[5]. Finally, the optimal transportation project is determined by combining other factors through hierarchical analysis method (AHP)[6]. Through the comprehensive use of multiple algorithms and models, this paper provides a comprehensive and in-depth analysis for the optimization of urban transportation system, which is an important reference value for urban transportation planning and decision-making.

II. Transportation Network Construction and Analysis

A. Data Description

1) Missing Value Handling

For missing values in the data channel and maxspeed, the data is completed using median padding, which is more reliable for the presence of outliers or skewed distributions.

2) Outlier Handling

Outliers are identified using IQR. The interquartile range is the difference between the upper and lower quartiles. By using 1.5 times the IQR as a criterion, the study designates points where $xi < Q1 - 1.5IQR$ or $xi > Q3 + 1.5IQR$ is an outlier. $q1$ is the 25th percentile of all sample values ordered from smallest to largest. $q3$ is the 75th percentile of all sample values ordered from smallest to largest. $q1$ is the 25th percentile of all sample values ordered from smallest to largest. $q3$ is the 75th percentile of all sample values ordered from smallest to largest.

$$IQR = Q3 - Q1 \tag{1}$$

3) Standardized Processing

AADT data were processed using the maximization process and the following equations:

$$x' = \frac{x_i}{\max(X)} \tag{2}$$

B. Directed Graph Modeling

1) Transportation Network Construction

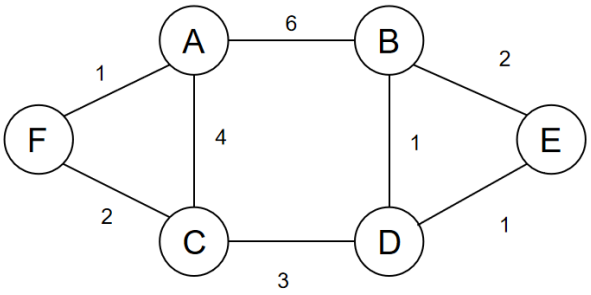


Fig. 1. Schematic of a weighted directed graph

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As shown in Fig. 1, the transportation system of a city is modeled as a weighted directed graph $G = (V, E)$: V: the set of nodes, which denotes the locations of intersections, bus stops, transportation hubs, etc. E: the set of edges, which denotes the road or traffic connections between nodes. Each edge eij is assigned a weight w_{ij} which is calculated from the combination of distance, traffic flow and number of lanes.

2) Construction of Network Weights

For each edge eij, the combination weight formula is:

$$w_{ij} = \frac{d_{ij}}{v_{ij}} + c_{ij} \tag{3}$$

Where: d_{ij} is the length of the road, v_{ij} is the average speed of vehicles traveling on that road, and c_{ij} is the congestion factor. w_{ij} is smaller, indicating a more efficient side.

3) Network Topology Attributes

The following key attributes of the network are calculated:

(a) Node degree:

$$Degree(\mathrm{i}) = \sum_{\mathrm{j} \in V} A_{\mathrm{ij}} \tag{4}$$

Where A_{vw} is an element of the adjacency matrix.

(b) Shortest path:

Using Dijkstra's algorithm, the shortest path d_{ij} is defined:

$$d_{ij} = \min w_{ij}, \forall e_{ij} \in P \tag{5}$$

(c) Degree of congestion:

$$C_{\mathrm{ij}} = \frac{1}{\sum_{\mathrm{i} \neq \mathrm{j}} d_{\mathrm{ij}}} \tag{6}$$

Reflects the transportation accessibility between nodes.

4) Impact Assessment after Bridge Collapse

After collapse, delete the node indicating bridge:

$$G' = \left(V - v_{birdge}, E - e_{birdge}\right) \tag{7}$$

The study will assess the impact based on the shortest path and traffic distribution before and after the bridge collapse.

Shortest Path Algorithm: Dijkstra's algorithm calculates the best path from each starting point to the destination:

$$\min \sum_{(i,j)} w_{ij} x_{ij} \min(i, j) \tag{8}$$

Where $x_{ij} \in 0,1$ denotes the path selection variable.

5) Redistribution After Bridge Collapse

The study is forced to minimize commuting time and maximize road utilization.

$$\min \sum_{e_{ij} \in E} w_{ij} \cdot x_{ij} \tag{9}$$

Where x_{ij} is the binary decision variable and $x_{ij} = 1$ denotes the selected edge.

Constraints are included:

$$f_{ij} \leq c_{ij}, \forall (ij) \in E \tag{10}$$

After the bridge is rebuilt, flows need to be redistributed and compared to what they were at the time of the collapse. The study chose to use a comparative analysis of the before and after network:

$$\Delta T = T_{collapsed} - T_{rebuilt} \tag{11}$$

Where $T_{collapsed}$ is the total commute time during the collapse and $T_{rebuilt}$ is the total commute time after rehabilitation.

C. Solving with Dijkstra's Algorithm

Based on the model constructed above, combined with a map of the city, it can be seen that there are two alternative paths to this bridge, a freeway and a tunnel. Thus the changes in traffic flow before and after the bridge collapse can be derived as shown in table I.

TABLE I. COMPARISON OF ANNUAL TRAFFIC FLOW BEFORE AND AFTER BRIDGE, TUNNEL AND HIGHWAY COLLAPSE

Annual traffic volume (10,000 vehicles)	Before the collapse	After the collapse
Highway	10367.998	10858.75
Tunnel	6846	6923
Bridge	1150	0

A comparison of traffic impacts for the remaining other stakeholders before and after the bridge collapse is shown in table II.

TABLE II. COMPARISON OF ANNUAL TRAFFIC BEFORE AND AFTER COLLAPSE FOR OTHER STAKEHOLDERS

Annual traffic volume (10,000 vehicles)	Before the collapse	After the collapse
airport1	21871	23532
airport2	9572	11180
The nearest transportation hub connected to the bridge	38453	36010
The nearest transportation hub connected to the tunnel	13093	6310
The largest transportation hub in the lower right corner	75343	76080
The largest transportation hub in the city center	22581	24302
The largest transportation hub connected with the northwest cities	54301	57452
The largest transportation hub connected with the northern cities	2702	2500

D. Summary Analysis

1) Impacts of Bridge Collapse

Significant increase in commuting time; soaring transportation costs; restriction of commercial activities; forced adjustment of travel plans; worse tourist experience; impediment to economic development and industrial upgrading of the city.

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2) Impacts of Bridge Reconstruction

(a) Impacts during construction

Increased traffic congestion in the vicinity: during the simulated construction period, the capacity of the surrounding roads will be reduced due to the occupation of roads by the construction site and traffic control.

Disruption of residents' lives: Construction noise and dust negatively affect the quality of life of nearby residents.

(b) Impacts after completion

Significant improvement in transportation: All stakeholders benefit from the reassessment of the transportation system after the bridge reconstruction is completed.

Increased attractiveness of the city: Improved transportation enhanced the image of the city and attracted more tourists to the city.

III. Optimization of Public Transportation Systems

A. Improvements to the Directed Graph Model

In this section, we choose to optimize the city's bus transportation system and continue to model the bus system using the weighted directed graph in the previous section $G = (V, E)$, where:

V denotes a node, i.e., a bus stop. E denotes an edge, i.e., a road.

The edge weight w_{ij} denotes the total cost spent on path passage, including passage time, passage distance, congestion level, etc., defined as follows:

$$w_{ij} = \alpha \cdot D' + \beta \cdot U' + \gamma \cdot K' \tag{12}$$

D denotes length data, U denotes AADT, and K denotes Lanes.

Bus stop coverage analysis: the coverage of bus stops is within walking distance, and here the coverage radius is defined as 500 meters.

The formula is as follows:

$$C(v_i) = \frac{|V_{covered}|}{|V|} \tag{13}$$

Where $V_{covered}$ denotes the set of nodes covered in range r.

Network-wide coverage: calculate the bus stop coverage of all nodes in the constructed directed graph.

$$C_{total} = \frac{\sum_i C(v_i)}{|V|} \tag{14}$$

Reachability metrics: the shortest path length d_i from node i to the nearest bus stop, the average reachability formula is as follows:

$$H = \frac{\sum_i d_i}{|V|} \tag{15}$$

Next, bus route optimization is performed by ant colony algorithm. The specific design of ACO algorithm is as follows.

State transfer rule: ants move between nodes in the transportation network and select the next node based on the pheromone concentration of edges and heuristic information. The heuristic information can be defined as the potential contribution of the bus lane to each objective. For example, for the efficiency of bus operation, the road that enables buses to avoid congested roads has a higher value of heuristic information. ant k transfers from node i to node j with probability:

$$p_{ij}^k = \frac{[\tau_{ij}]^\alpha [\eta_{ij}]^\beta}{\sum_{l \in allowed_k} [\tau_{il}]^\alpha [\eta_{il}]^\beta} \tag{16}$$

Where τ_{ij} is the pheromone concentration on edge (i,j), η_{ij} is the heuristic information, α and β are the relative importance of the pheromone and the heuristic information, respectively, and allowedk is the set of nodes that can be chosen by ant k next.

Pheromone update rule: all ants update the pheromone after completing one path search. The pheromone volatilization formula is :

$$\tau_{ij} = (1-\rho)\tau_{ij} \tag{17}$$

Where ρ is the pheromone volatilization rate, $0 < \rho < 1$ Meanwhile, the pheromone enhancement is performed according to the contribution of the paths traveled by the ants to the objective function. If the value of the objective function corresponding to the path traveled by ant k is Fk , the pheromone enhancement is:

$$\Delta\tau_{ij}^k = \frac{Q}{F_k} \tag{18}$$

Where Q is a constant. The updated pheromone is:

$$\tau_{ij} = \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^k \tag{19}$$

M is the number of ants.

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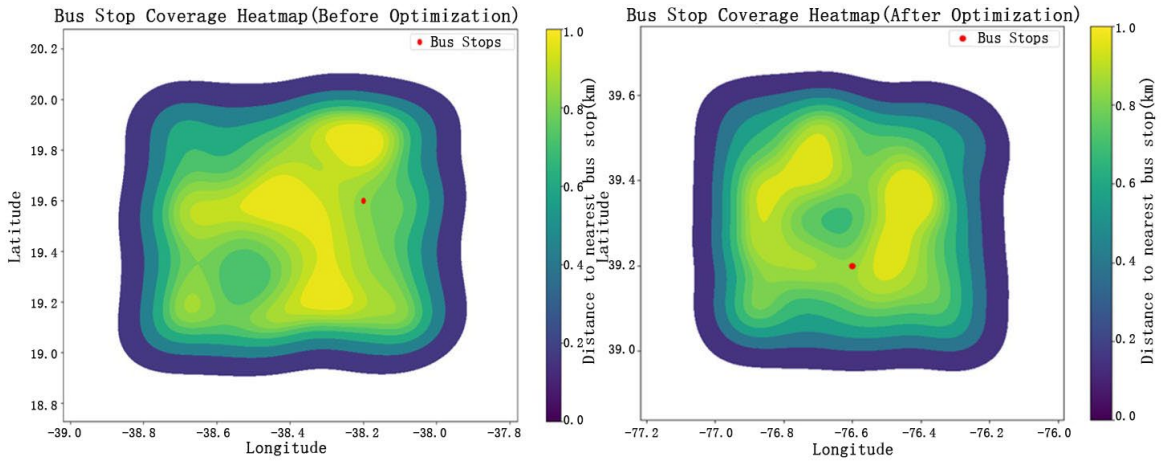


Fig. 2. Bus stop coverage before and after optimization

B. Solving with Ant Colony Algorithm

Initialize the number of ants, pheromone concentration, parameters α , β , ρ etc. Let the ants randomly choose the starting node in the transportation network and construct the paths according to the state transfer rule. Calculate the value of the objective function corresponding to each path and update the pheromone. Repeat the above process until the iteration termination condition is satisfied (e.g., the objective function value converges, or the maximum number of iterations is reached).

The final heat map derivation is shown in Fig. 2.

After optimization, the bus stop coverage increased from 60% to 70% and the average accessibility was reduced from 1.5 km to 1.2 km.

IV. Optimal Transportation Project Evaluation

A. Establish a model by comprehensively using AHP and entropy method

This section entails identifying a transportation network project that will best improve the lives of the people in the city, and examines the use of hierarchical analysis and combining the data with the more objective entropy method to assign values.

Layered analysis is a decision analysis method that combines qualitative and quantitative methods for solving complex situations with multiple objectives. The method combines quantitative and qualitative analysis by using the decision maker's experience to determine the relative importance of the criteria that measure whether or not the objectives can be achieved, and reasonably assigns a weight to each criterion for each decision scenario, and uses the weights to find the order of strengths and weaknesses of each scenario. For this section there are three indicators that need to be quantified, so consider using hierarchical analysis to solve it.

Hierarchical analysis is mainly divided into the following steps: establishing a multilevel structural model, constructing a two-by-two comparison judgment matrix, weight calculation, consistency test, and overall importance calculation.

1) Establishment of a multi-level ladder structure model

According to the advantageous relationship, the evaluation index system is established in three layers from top to bottom:

(a) Top Layer: also known as Purpose Layer or Goal Layer, it is the goal or result that the system wants to achieve, and it is the primary criterion for system evaluation.

(b) Criteria Layer: Criteria, sub-criteria, etc. established to realize the goal layer.

(c) Bottom Layer: also known as Program Layer. It is the various programs, measures, etc. taken to achieve the objectives.

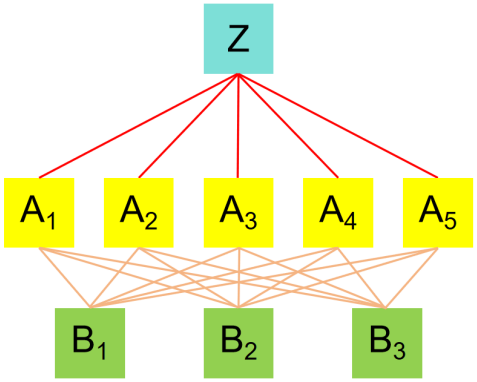


Fig. 3. Structural Modeling of Analytic Hierarchy Process

As shown in Fig. 3, Z is the decision layer $A1,\ldots..A5$ is the criterion layer and $B1$, $B2$ and $B3$ are the bottom layer (program layer).

For this section the objective layer is to maximize the improvement of people's livelihood, the criterion layer is traffic efficiency, traffic comfort, impact on the surrounding environment, and residents' satisfaction, etc., and the program layer is the optimization of the public transportation system, the optimization of the private car system, the optimization of the walking system, and the optimization of other travel modes.

(d) Use entropy value method to construct 2×2 comparison judgment matrix and weight calculation.

Normalize the factors by the number of each option.

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TABLE III. CONSISTENCY TEST RESULTS

	Tansportation efficiency	Impact on the Surrounding environment	Resident Satisfaction level	Transportation comfort	Eigenvector (Math.)	Weighted value (%)	CR value	Consistency test
Transportation efficiency	1	0.333	0.2	5	0.76	12.689	0.037	pass
Impact on the surrounding environment	3	1	0.2	5	1.316	21.979	0.037	pass
Resident satisfaction level	5	5	1	7	3.637	60.741	0.037	pass
Transportation comfort	0.2	0.2	0.143	1	0.275	4.592	0	pass

(e) Due to the existence of the platform, "positive indicators" and "negative indicators", these two types of data will be pre-processed separately. The reason for subtracting 0.0001 from the minimum value and adding 0.0001 to the maximum value is that the overall effect on the fruit is not significant and can be ignored if the whole column is the same value.

$$X_{\min} = \min(x_{1j}, x_{2j}, \ldots, X_{nj}) - 0.0001 \tag{20}$$

$$\mathrm{X}_{\max} = \max(x_{1j}, x_{2j}, \ldots x_{nj}) + 0.0001 \tag{21}$$

For positive indicators:

$$Z_{ij} = \frac{X_{ij} - X_{\min}}{X_{\max} - X_{\min}} \tag{22}$$

For negative indicators:

$$Z_{ij} = \frac{X_{\max} - X_{ij}}{X_{\max} - X_{\min}} \tag{23}$$

Calculate the entropy of the *J*th indicator:

$$e_j = -k\sum i = \ln pij \ln(p_{ij}), j = 1, \ldots, m \tag{24}$$

Calculate the information entropy redundancy (variance):

$$d_j = 1 - e_j, j = 1, \ldots, m \tag{25}$$

Calculate the weights of the indicators:

$$W_j = \frac{d_j}{\sum_{j=1}^{m} d_j}, j = 1, \ldots, m \tag{26}$$

Calculate the composite score for each sample:

$$S_i = \sum_{j=1}^{m} \mathrm{w}_j x_{ij}, i = 1, \ldots, n \tag{27}$$

Where x_{ij} is the normalized data. Based on the scores of each influence factor, the importance ranking of all factors can be derived. The weight of transportation efficiency is 12.689%, the weight of influence on the surrounding environment is 21.979%, the weight of residents' satisfaction is 60.741%, and the weight of transportation comfort is 4.592%.

(6) Consistency test

In order to ensure the correctness and reasonableness of the obtained weights, a consistency test is also required.

The consistency index $C.I.$ is calculated:

$$C.T. = \frac{\lambda_{\max} - n}{n - 1} \tag{28}$$

Where.

$$\lambda_{\max} = \sum \frac{(AW)_i}{nW_i} \tag{29}$$

Obviously, the larger n is, the larger the error in the $C.I.$ will be. Therefore, a random consistency ratio $C.R.$ is introduced in the test.

$$C.R. = \frac{C.I.}{C.R.} \tag{30}$$

When the random consistency ratio $C.R. < 0.1$, it is considered that the calculated hierarchical ranking weights are correct and reasonable, otherwise the judgment matrix needs to be readjusted until the consistency test is qualified.

The consistency test results require a CR value ($CR = CI / RI$) of less than 0.1, which is used to determine whether there is a logical error when one constructs the judgment matrix, e.g., there are three indicators ABC, and judgment A is more important than B, and B is more important than C. The result of the consistency test is that A is more important than B, and B is more important than C. Therefore, logically, A must be more important than C. However, if A is more important than C when constructing the judgment, then A must be more important than C. However, if A is more important than C when constructing the judgment, and it is assumed that C is more important than A, then a logical error has been made and the consistency test cannot be passed. The calculation results of the hierarchical analysis method in this section show that according to the RI table, the maximum characteristic root is 4.364, and the corresponding RI value is 0.882, so $CR = CI/RI = 0.037 < 0.1$, which passes the one-time test. The results are summarized in table III.

B. Solving with AHP

The option with the highest weight is the best choice to achieve the goal, and the other options are ranked in order according to their weights, with the larger weighted options being better and the smaller weighted options being worse. The total ranking of the program hierarchy in this section is shown in table IV.

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TABLE IV. SUMMARY RESULTS OF SCHEME LEVEL JUDGMENT MATRIX BY SCHEME

Nodal Item	Optimization of the public transport system	Optimization of the private car system	Optimization of the pedestrian system
Transportation efficiency	0.258	0.637	0.105
Impact on the surrounding environment	0.258	0.105	0.637
Resident satisfaction level	0.637	0.258	0.105
Transportation comfort	0.2	0.6	0.2

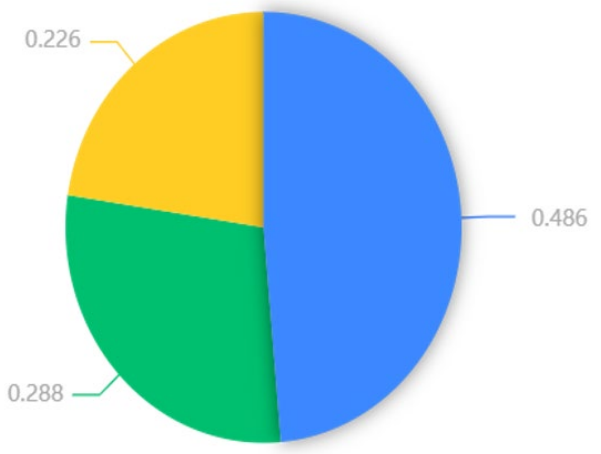


Fig. 4. The scoring graph for each scenario

As shown in Fig. 4, the final result is a scenario optimization score of 0.486 for the public transportation system, 0.288 for the private vehicle system, and 0.226 for the pedestrian system.

V. CONCLUSION

In this paper, we conduct an in-depth study on the situation that exists in the urban transportation system by integrating the use of computer-related models such as transportation network modeling and directed graphical modeling, as well as Dijkstra's algorithm, Ant Colony Algorithm (ACO), and Hierarchical Analysis (AHP) combined with the entropy method. Data quality was improved through data preprocessing, a weighted directed graph was constructed to describe the transportation system, the impact of bridge collapse was evaluated, bus routes were optimized, and the optimized public transportation system was identified as the best transportation project. The study shows that bridge collapses can seriously affect transportation and related stakeholders, and transportation significantly improves after reconstruction. Optimizing bus routes and stops can effectively alleviate the traffic situation. These research results provide a scientific basis for urban transportation planning and decision-making, and have important practical application value.

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